Table 7.2 Atomic mass of some elements

Atomic Number	Name	Symbol	Atomic Mass (amu)
1	Hydrogen	H	1.008
2	Helium	He	4.003
3	Lithium	Li	6.941
4	Beryllium	Be	9.012
5	Boron	B	10.811

Calculation of average atomic mass – Solved Examples

Example 1: Oxygen is the most abundant element in both the Earth's crust and the human body. It exists as a mixture of three stable isotopes in nature as shown in Table 7.3:

Table 7.3 Isotopes of oxygen

Isotope	Mass (amu)	% abundance
$_8O^{16}$	15.9949	99.757
$_8O^{17}$	16.9991	0.038
$_8O^{18}$	17.9992	0.205

The atomic mass of

oxygen = $(15.9949 \times 0.99757) + (16.9991 \times 0.00038) + (17.9992 \times 0.00205)$

= 15.999 amu.

Example 2: Boron naturally occurs as a mixture of boron-10 (5 protons + 5 neutrons) and boron-11 (5 protons + 6 neutrons) isotopes. The percentage abundance of B-10 is 20 and that of B-11 is 80. Then, the atomic mass of boron is calculated as follows:

Atomic mass of

$$\text{boron} = \left(10 \times \frac{20}{100}\right) + \left(11 \times \frac{80}{100}\right)$$

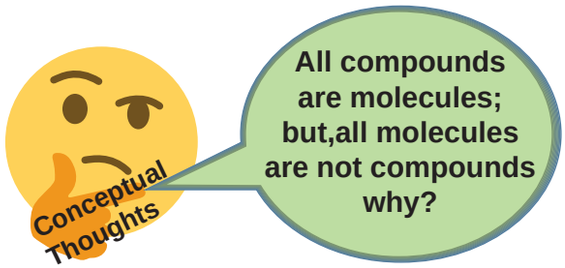
$= (10 \times 0.20) + (11 \times 0.80)$

$= 2 + 8.8$

= 10.8 amu

7.2 MOLECULE AND MOLECULAR MASS

Except noble gases, atoms of most of the elements are found in the combined form with itself or atoms of other elements. It is called as a molecule. **A molecule is a combination of two or more atoms held together by strong chemical forces of attraction, i.e. chemical bonds.**



7.2.1 Classification of molecules

A molecule may contain atoms of the same element or may contain atoms of two or more elements joined in a fixed ratio, in accordance with the law of definite proportions. Thus, a molecule may be an **element or a compound**. If the molecule is made of similar kind of atoms, then it is called **homoatomic molecule**.

The molecule that consist of atoms of different elements is called **heteroatomic molecule.** A compound is a heteroatomic molecule. The number of atoms present in the molecule is called its '**atomicity**'.

Table 7.4 Classification of molecules

Atomicity	No. of atoms present	Name
1	1	Monoatomic
2	2	Diatomic
3	3	Triatomic
>3	>3	Polyatomic

Let us consider oxygen. Oxygen gas exists in two allotropic forms: Oxygen (O_2) and Ozone (O_3). In oxygen molecule, there are two oxygen atoms. So its atomicity is two. Since both the atoms are similar, oxygen (O_2) is a homodiatomic molecule. Other elements

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